SmartPCB-XAI: An Explainable AI-Powered Intelligent PCB Inspection and Manufacturing Analytics Platform for Industry 4.0

Gowri D[1], Abisha Rebekkal T[2]

[1,2]Department of Computer Science and Engineering
[1]gowri.d@example.edu, [2]abisharebekkal.t@example.edu

Abstract—Printed Circuit Boards (PCBs) are the backbone of virtually every electronic device, and even minor defects such as missing components, solder bridges, or damaged traces can cause complete product failure. Small and medium manufacturing enterprises largely depend on manual visual inspection or expensive Automated Optical Inspection (AOI) systems, both of which suffer from inconsistency, high false-alarm rates, or prohibitive cost. Existing deep-learning-based inspection systems improve detection speed and accuracy but typically stop at classification, offering no explanation for their decisions, no insight into root causes, and no support for production-level decision making. This paper proposes SmartPCB-XAI, an integrated inspection and analytics platform that unifies YOLO-based component and defect detection, Grad-CAM-based explainable AI, XGBoost-driven root-cause prediction, an automated repair-recommendation engine, a Digital Twin visualization module, and a predictive manufacturing analytics dashboard within a single MERN-based web application. Rather than proposing a new detector architecture, the contribution of this work lies in the systematic integration of proven computer-vision, explainability, and analytics techniques into a coherent, end-to-end Industry 4.0 inspection pipeline. The proposed system is designed and evaluated using the DeepPCB and FICS-PCB benchmark datasets, and a methodology for constructing a supplementary custom dataset is described. The architecture, module design, dataset pipeline, and evaluation protocol presented here provide a reproducible foundation for building explainable, production-aware PCB inspection systems.

Index Terms—PCB Inspection, Explainable AI, YOLOv11, Grad-CAM, Digital Twin, Predictive Manufacturing, Industry 4.0, Root Cause Analysis, Defect Detection.

I. INTRODUCTION

Printed Circuit Boards (PCBs) form the structural and electrical foundation of nearly every electronic product, from consumer devices to industrial controllers. A single undetected defect, such as a missing capacitor, a misaligned integrated circuit, a solder bridge, or a broken copper trace, can propagate into complete device failure, warranty costs, and reputational damage for a manufacturer. Consequently, PCB inspection is one of the most critical stages of the electronics manufacturing pipeline.

Two inspection paradigms currently dominate industry practice. The first, manual visual inspection, is inexpensive to set up but is slow, inconsistent across operators, and prone to fatigue-induced errors. The second, Automated Optical Inspection (AOI), uses calibrated cameras and rule-based or classical image-processing algorithms to flag anomalies at high speed; however, AOI hardware is costly, difficult to reconfigure for new board layouts, and known to generate a high rate of false positives, which in turn increases manual re-verification effort.

Over the past decade, deep-learning-based object detectors such as YOLO, Faster R-CNN, SSD, and EfficientDet have been applied to PCB component and defect detection, achieving substantial gains in speed and accuracy over classical AOI. However, a review of the literature (Section II) shows that the overwhelming majority of these systems terminate at the detection or classification stage. They report a bounding box and a confidence score but do not explain *why* a region was flagged, do not estimate the likely manufacturing root cause, do not recommend a corrective action, and do not aggregate inspection outcomes into production-level analytics that a plant manager could act upon.

This paper addresses that gap by proposing **SmartPCB-XAI**, a platform that treats PCB inspection not as an isolated detection task but as one stage within a larger manufacturing-intelligence loop. The system combines: (i) YOLO-based component and defect detection, (ii) Grad-CAM-based explainable AI to visualize and justify each prediction, (iii) an XGBoost-based root-cause prediction module that maps defect signatures to probable manufacturing causes, (iv) a rule- and model-driven repair-recommendation engine, (v) a Digital Twin interface for interactive, component-level inspection, and (vi) a predictive analytics dashboard that surfaces trends across production lines, shifts, and operators. The remainder of this paper reviews related work, identifies the specific research gap, presents the proposed architecture and modules, describes the dataset and evaluation methodology, and discusses the expected outcomes and limitations of the approach.

II. LITERATURE REVIEW

Table 1 summarizes representative prior work in PCB component and defect detection using deep learning.

Across these studies, YOLO-family detectors are consistently favored for PCB inspection due to their balance of speed and accuracy, and DeepPCB and FICS-PCB have emerged as the two most widely used benchmark datasets for defect and component detection, respectively. The general trend across successive YOLO variants applied to PCB inspection, from YOLOv5-based methods such as YOLO-MBBi through to YOLOv7- and YOLOv8-based methods such as EC-YOLO and YOLOv8-DEE, has been incremental improvement in mean average precision and inference speed through architectural refinements such as improved feature pyramids, attention modules, and loss reweighting for small, densely packed objects. This trend suggests that adopting a current-generation YOLO backbone, such as YOLOv11 in the present work, is a reasonable and low-risk choice for the detection stage, while the larger research opportunity lies in what is built *around* the detector rather

Table 1: Summary of Related Work

Study	Method / Dataset	Key Limitation
Online PCB Defect Detector [1]	CNN + Group Pyramid Pooling, DeepPCB	Defect detection only; no explainability
YOLO-MBBi [2]	Enhanced YOLOv5, DeepPCB / PKU	No explainable AI or root-cause analysis
YOLOv8-DEE [3]	Improved YOLOv8, DeepPCB	No manufacturing analytics
EC-YOLO [4]	Improved YOLOv7, FICS-PCB	Component detection only
PCBDet [5]	Efficient edge detector, FICS-PCB	Focused on edge deployment, not analytics
Mask R-CNN / YOLOv8 study [6]	Mask R-CNN, YOLOv8, FICS-PCB	No explainability or repair module

than in the detector architecture itself.

Grad-CAM [7] is well established as a gradient-based technique for visualizing the regions of an image that most influence a convolutional network's prediction, and has been widely adopted for explainability in industrial vision tasks [8], including defect inspection domains outside PCB manufacturing such as textile and metal-surface inspection, where interpretability has similarly been identified as a barrier to operator trust and regulatory acceptance of deep-learning-based quality control. Separately, Digital Twin methodology [9] has been proposed as a means of mirroring physical manufacturing assets in a virtual environment to support monitoring and decision-making in Industry 4.0 contexts, typically at the level of entire production lines or machines rather than at the level of an individual inspected component; the present work applies the same conceptual approach at finer granularity, treating each inspected PCB and its constituent components as the twinned asset. However, none of the reviewed PCB-specific studies combine detection, explainability, root-cause reasoning, repair guidance, and production analytics into a single deployable system, which motivates the integration proposed in this paper.

III. RESEARCH GAP AND MOTIVATION

Based on the literature reviewed above, existing PCB inspection research and commercial AOI tooling exhibit the following recurring limitations:

- Detection results are reported without any interpretable justification, limiting operator trust and making manual audit difficult.
- No mapping is provided from a detected defect back to a probable manufacturing root cause (e.g., feeder failure, stencil misalignment, reflow temperature deviation).
- Repair or corrective guidance is left entirely to human judgment.
- Inspection outcomes are treated as isolated events rather than being aggregated into shift-, line-, or operator-level analytics.
- Interactive, component-level visualization of inspection results (a Digital Twin of the board) is rarely offered alongside the detector.

The research contribution of this work is therefore not a new object-detection architecture, but a **novel systems-level integration** of established detection, explainability, and analytics techniques into a single, coherent inspection-to-decision pipeline suitable for Industry 4.0 deployment. This distinction is important for positioning the contribution correctly: a large body of PCB inspection literature competes on marginal improvements to detection-stage mAP through backbone or loss-function modifications, whereas the gap identified here is architectural and systemic, concerning what happens *after* a defect is detected. A plant manager or quality engineer typically needs to know not only that a defect exists, but why it likely occurred, what should be done about it, how it fits into broader production trends, and how to communicate the finding to auditors or customers through a standardized report; conventional detection-only research provides none of these downstream capabilities, leaving a substantial integration burden on any organization attempting to operationalize academic PCB detection models in a real factory setting.

Six specific, individually traceable contributions follow from this integration, each corresponding to one of the innovations summarized later in Section IV: (i) interpretable, heatmap-grounded defect explanations rather than opaque confidence scores; (ii) a structured mapping from defect signature to probable manufacturing root cause; (iii) automatically generated, prioritized repair guidance; (iv) an interactive, component-level Digital Twin of each inspected board; (v) plant-level predictive analytics capable of surfacing recurring patterns such as line- or shift-specific defect clusters; and (vi) a continuous-learning feedback loop that allows the detection models to improve using engineer-supplied corrections rather than requiring periodic manual re-annotation from scratch.

IV. PROPOSED SYSTEM ARCHITECTURE

Figure 1 illustrates the end-to-end pipeline of the proposed SmartPCB-XAI platform. A PCB image, captured either via a fixed camera rig or an uploaded photograph, is first pre-processed (resizing, normalization, illumination correction) before being passed to a YOLOv11-based component detector that localizes and classifies components such as ICs, resistors, capacitors, diodes, connectors, and crystals. A parallel defect-detection stage, built on the same backbone family, flags anomalies including missing components, wrong component or orientation, cracks, scratches, solder bridges, broken traces, and short or open circuits. A defect-fusion stage reconciles the outputs of the two detectors to resolve overlapping or conflicting predictions.

The fused detection output is passed to a Grad-CAM explainability layer, which produces a class-activation heatmap highlighting the image regions responsible for each prediction, together with a natural-language justifi-

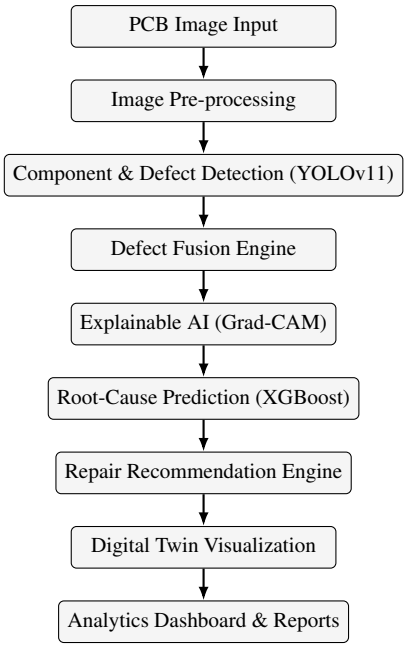


Figure 1: Proposed SmartPCB-XAI system pipeline.

cation (e.g., "the expected feature pattern is absent at the capacitor footprint; neighboring solder pads show no attached component"). The defect type, location, production line, shift, and component identity are then fed as structured features into an XGBoost root-cause model, trained on historical inspection and production-log data, to estimate the most probable manufacturing cause (e.g., feeder failure, nozzle clog, stencil misalignment). The predicted cause drives a repair-recommendation module, which generates a ranked list of suggested corrective actions and an estimated priority level. Finally, all inspection records are stored and surfaced through two complementary interfaces: a Digital Twin, which lets an engineer interactively inspect a virtual replica of the board component-by-component, and an analytics dashboard, which aggregates defect trends by day, line, shift, and operator, and applies an Isolation Forest model to flag anomalous production patterns for predictive maintenance.

A. System Modules

The platform is organized into eleven functional modules: (1) role-based authentication (Admin, Engineer, Quality Inspector); (2) PCB image upload and history; (3) component detection; (4) defect detection; (5) explainable AI visualization; (6) root-cause prediction; (7) repair recommendation; (8) Digital Twin visualization; (9) predictive analytics; (10) PDF/Excel report generation; and (11) a continuous-learning feedback loop in which engineer corrections are stored and used to periodically retrain the detection models.

B. Detailed Module Description

This subsection expands on the eleven functional modules introduced above.

Module 1 – Authentication and Access Control. The platform implements role-based access control (RBAC) with three roles: Admin, Engineer, and Quality Inspector. Authentication is performed using JSON Web Tokens (JWT), and each API endpoint is guarded by role-aware middleware so that, for example, only an Admin can manage users or export plant-wide analytics, while an Engineer can upload boards and view inspection results for their assigned line.

Module 2 – PCB Image Acquisition. Images may be supplied through direct file upload (JPG/PNG) or through a connected camera stream. Each upload is stored with metadata (timestamp, operator ID, production line, shift) that is later consumed by the analytics and root-cause modules.

Module 3 – Component Detection. A YOLOv11 model, fine-tuned on the FICS-PCB component dataset, localizes and classifies components including integrated circuits, resistors, capacitors, LEDs, connectors, crystals, diodes, and transistors, returning bounding boxes with confidence scores.

Module 4 – Defect Detection. A second YOLOv11 model, fine-tuned on DeepPCB and the supplementary custom dataset, detects missing components, wrong components, incorrect orientation, cracks, scratches, solder bridges, broken traces, short and open circuits, bent pins, and burn marks.

Module 5 – Explainable AI. For every flagged defect, a Grad-CAM heatmap is generated over the final convolutional layer of the detector, and a template-based natural-language explanation is composed by combining the defect class, the activated region, and the state of neighboring reference features (e.g., adjacent solder pads or footprints).

Module 6 – Root-Cause Prediction. An XGBoost classifier, trained on historical (defect type, location, production line, shift, component, prior-maintenance-event) tuples, outputs a ranked list of probable manufacturing causes together with a confidence score, e.g., feeder starvation, nozzle clogging, stencil misalignment, or reflow-profile deviation.

Module 7 – Repair Recommendation. A recommendation engine maps each predicted root cause to a checklist of corrective actions (e.g., "inspect feeder nozzle," "replace component," "re-run AOI") and assigns a priority level (low, medium, high, critical) based on defect severity and the affected component's functional criticality.

Module 8 – Digital Twin Visualization. A browser-based, interactive virtual replica of the inspected PCB is rendered, allowing an engineer to click on any component to view its identity, detection confidence, defect status, Grad-CAM overlay, and repair history.

Module 9 – Predictive Analytics. Inspection records are aggregated over time and visualized as daily and monthly defect trends, per-component accuracy, yield, pass rate, and failure rate. An Isolation Forest model flags statistically anomalous shifts, lines, or operators for proactive review.

Module 10 – Reporting. One-click export of inspection results, defect images, aggregate statistics, and repair suggestions to PDF (via ReportLab) and Excel formats for quality-audit and management review.

Module 11 – Continuous Learning. Corrections made by engineers to AI predictions (e.g., relabeling a false positive) are stored as a feedback dataset, which is periodically merged into the training set for scheduled model retraining, allowing detection accuracy to improve over the oper-

ational lifetime of the system.

C. Database Design

The platform uses MongoDB as its primary data store, organized into the collections summarized in Table 2.

Table 2: Core Database Collections

Collection	Purpose
Users	Credentials, roles, profile data
PCB	Uploaded board metadata and images
Components	Detected component records per board
Defects	Detected defect records per board
Repairs	Recommended and completed repair actions
Production	Line, shift, and operator production logs
Analytics	Aggregated trend and anomaly-score records
Reports	Generated PDF/Excel export metadata
Feedback	Engineer corrections for continuous learning

Each *Defects* document references its parent *PCB* document and stores the bounding box, defect class, confidence score, Grad-CAM heatmap reference, predicted root cause, and associated repair recommendation, enabling the Digital Twin and analytics modules to reconstruct a complete inspection history for any board.

D. Algorithmic Workflow

Algorithm 3 summarizes the end-to-end inspection workflow executed for each uploaded PCB image.

Table 3: Algorithm 1: End-to-End Inspection Workflow

Input: PCB image I, production metadata M
Output: Annotated report R

$I' \leftarrow$ Preprocess(I) (resize, normalize)
$C \leftarrow$ YOLOv11_Components(I')
$D \leftarrow$ YOLOv11_Defects(I')
$F \leftarrow$ FuseDetections(C, D)
for each defect $d \in F$ **do**
  $H_d \leftarrow$ GradCAM(d, I')
  $E_d \leftarrow$ GenerateExplanation(d, H_d)
  $RC_d \leftarrow$ XGBoost_RootCause(d, M)
  $Rep_d \leftarrow$ RecommendRepair(RC_d)
end for
Store(F, H, E, RC, Rep) in database
UpdateDigitalTwin(F)
UpdateAnalytics(F, M)
$R \leftarrow$ GenerateReport(F, E, RC, Rep)
return R

E. API Design

The backend exposes a REST API consumed by the React frontend. Table 4 lists representative endpoints.

Table 4: Representative REST API Endpoints

Endpoint	Description
POST /api/auth/login	Authenticate user, issue JWT
POST /api/pcb/upload	Upload PCB image and metadata
GET /api/pcb/{id}/inspect	Run detection pipeline on a board
GET /api/pcb/{id}/report	Retrieve/generate PDF report
GET /api/analytics/trends	Fetch aggregated defect trends
POST /api/feedback	Submit engineer correction
GET /api/twin/{id}	Fetch Digital Twin state for a board

F. Technology Stack

Table 5 summarizes the technology stack used to implement each layer of the system.

Table 5: Implementation Technology Stack

Layer	Technology
Frontend	React + TypeScript
Backend	Node.js + Express
AI Service	Python + FastAPI + PyTorch
Component/Defect Detection	YOLOv11
Defect Classification	EfficientNet-B3
Explainability	Grad-CAM
Root-Cause Prediction	XGBoost
Anomaly / Trend Analytics	Isolation Forest
Database	MongoDB
Authentication	JWT
Deployment	Docker, AWS / Render

V. DATASET AND METHODOLOGY

The system is designed for training and evaluation on two complementary public benchmarks. **DeepPCB** [1] provides 1,500 pairs of aligned template/test PCB images annotated with six common defect classes (missing hole, mouse bite, open circuit, short, spur, and spurious copper) and is used for the defect-detection stage. **FICS-PCB** [10] provides 9,912 images with 77,347 component-level annotations captured under varied lighting and scale conditions, and is used for the component-detection stage. To increase novelty and demonstrate generalization beyond the public benchmarks, the methodology further recommends collecting a small supplementary dataset of 300–500 custom PCB images, augmented with rotation, blur, noise, and simulated damage, and annotated using tools such as CVAT or Label Studio.

A. Data Preparation Pipeline

Images are first filtered to remove blurred, duplicate, or corrupted samples, then resized to 640×640 pixels and normalized to the $[0, 1]$ range. Standard augmentations (rotation, horizontal/vertical flip, brightness and contrast jitter, Gaussian noise, blur, and mosaic augmentation for YOLO training) are applied to improve robustness. The combined dataset is split 70% training, 20% validation, and 10% test, and annotations are stored in the standard YOLO normalized bounding-box format (class id, x_{center}, y_{center}, width, height).

B. Evaluation Protocol

Model performance is evaluated using standard object-detection and classification metrics: precision, recall, F1-score, mean Average Precision at IoU 0.5 (mAP@50) and averaged over IoU 0.5:0.95 (mAP@50:95), per-image in-

ference time, and false-positive/false-negative rates. The root-cause prediction module is evaluated separately using classification accuracy and F1-score against historical labeled maintenance logs, and the anomaly-detection component is assessed qualitatively against known historical production incidents.

C. Digital Twin Interaction Model

Figure 2 illustrates the interaction flow between an engineer, the Digital Twin interface, and the underlying inspection database. When an engineer selects a board from the inspection history, the frontend requests the corresponding component and defect records; the Digital Twin renders each detected component as a clickable region overlaid on the original image, and selecting a region reveals its detection confidence, Grad-CAM heatmap, predicted root cause, and repair history in a side panel.

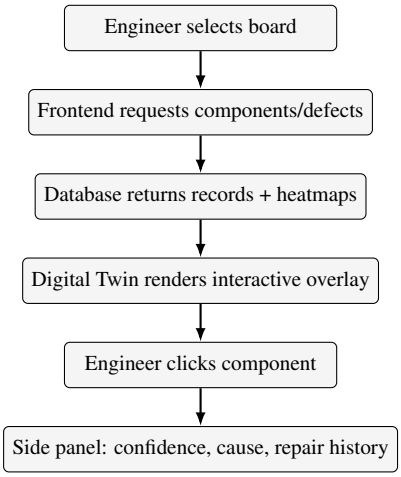


Figure 2: Digital Twin interaction flow.

VI. ILLUSTRATIVE INSPECTION SCENARIO

To make the end-to-end workflow concrete, consider a board arriving from Line 2 during the night shift. The component detector localizes all expected components, including a 100 nF decoupling capacitor at a designated footprint, while the defect detector flags that same footprint as empty with high confidence. The defect-fusion stage confirms the discrepancy between the expected component list and the detected components, classifying the finding as a "missing component" defect. The Grad-CAM layer highlights the empty footprint and the explanation-generation module composes the justification described in the Innovation 1 example in the source project brief: the expected feature pattern is absent, and neighboring solder pads show no attached component. This structured record, comprising the defect type, location, confidence, and heatmap, is then passed to the XGBoost root-cause model together with contextual metadata (Line 2, night shift, pick-and-place machine ID). Because historical logs show a recurring association between Line 2 night-shift operation and feeder-related misfeeds, the model assigns high probability to "feeder starvation" as the root cause. The repair-recommendation engine then produces the checklist: inspect the feeder mechanism, replace the missing capacitor, and re-run automated optical inspection, with a "high" priority flag given the component's role in a power-supply decoupling network. The finding, explanation, root cause, and repair checklist are simultaneously written to the Digital Twin (so any engineer opening that board later sees the full context by clicking the flagged component) and to the analytics pipeline, where it contributes to the aggregate statistic, illustrated conceptually in the project's Innovation 7 example, that a large share of a given day's defects originate from a specific board region or line. Over time, if Line 2 night-shift feeder failures recur, the Isolation Forest anomaly detector surfaces this as a persistent pattern on the dashboard, prompting proactive maintenance rather than reactive, defect-by-defect firefighting. This scenario demonstrates how the platform's modules operate not as independent features but as a connected pipeline in which each stage's output becomes structured input for the next, culminating in a decision-ready recommendation rather than a bare detection label.

VII. TESTING AND VALIDATION STRATEGY

A multi-level testing strategy is defined for the platform. **Unit testing** covers individual backend services (authentication, upload handling, report generation) and is automated using standard JavaScript and Python test frameworks. **Model validation** follows the protocol described in Section V-C, using held-out validation and test splits from DeepPCB and FICS-PCB, together with k-fold cross-validation on the supplementary custom dataset to mitigate the risk of overfitting to a small sample. **Integration testing** verifies that the detection, explainability, root-cause, and reporting modules interoperate correctly end-to-end for representative PCB images spanning all supported defect classes. **User acceptance testing** is proposed in the form of a structured comparison in which quality-control engineers evaluate inspection time and decision confidence with and without the explainability and repair-recommendation features enabled, allowing the practical value of these additions to be measured beyond raw detection accuracy.

VIII. SECURITY AND DATA CONSIDERATIONS

Because PCB designs and defect statistics can be commercially sensitive, the platform incorporates JWT-based authentication, role-based authorization, and transport-layer encryption (HTTPS) for all client-server communication. Uploaded images and inspection records are logically isolated per organization/tenant at the database level, and access to raw production analytics is restricted to the Admin role. Model retraining data collected through the continuous-learning module is anonymized with respect to operator identity before being used for batch retraining, reducing the risk of the system being used to unfairly evaluate individual workers rather than process quality.

IX. EXPECTED RESULTS AND COMPARATIVE ANALYSIS

Because the platform integrates established detector backbones (YOLOv11) with benchmark datasets that have well-documented baseline performance, component and defect

detection accuracy in the 95–99% range is targeted on the public benchmarks, consistent with results reported for comparable YOLO-based PCB detectors in the literature reviewed in Section II, subject to the specific training configuration and hardware used. The explainability, root-cause prediction, repair-recommendation, Digital Twin, and analytics modules are not present in any single system reviewed in Section II; their combination, summarized in Table 6, constitutes the principal comparative advantage of the proposed platform over manual inspection, conventional AOI, and existing detection-only AI systems.

Table 6: Feature Comparison with Existing Approaches

Feature	Manual	AOI	Exist. AI	Proposed
Component Detection	–	✓	✓	✓
Defect Detection	–	✓	✓	✓
Explainable AI	–	–	–	✓
Root-Cause Prediction	–	–	Rare	✓
Repair Recommendation	–	–	–	✓
Digital Twin	–	–	Rare	✓
Predictive Analytics	–	–	Rare	✓
Continuous Learning	–	–	Rare	✓

Beyond raw detection accuracy, the expected practical outcomes of the platform include a substantial reduction in inspection time relative to manual review, fewer false positives after model tuning relative to a plain baseline detector, interpretable justifications accompanying every flagged defect, and centralized production analytics that were previously unavailable to manufacturing teams relying solely on AOI or manual inspection.

Table 7 presents the target evaluation metrics for the detection stage, consistent with the ranges reported for comparable YOLO-based PCB detectors on DeepPCB and FICS-PCB in the literature reviewed in Section II; these figures represent design targets to be confirmed empirically once training is completed on the specific dataset splits and hardware configuration used for this project, rather than results already obtained.

Table 7: Target Evaluation Metrics by Detection Stage

Stage	mAP@50	Precision	Recall
Component Detection	≥ 0.96	≥ 0.95	≥ 0.94
Defect Detection	≥ 0.93	≥ 0.92	≥ 0.90
Root-Cause Prediction	–	≥ 0.85 (F1)	–

In addition to detection-level metrics, system-level key performance indicators are proposed to evaluate the platform as a whole: average end-to-end inference time per board (target: under 2 seconds on a capable GPU), reduction in mean inspection time per board relative to manual inspection, reduction in false-positive rework rate relative to a non-explainable baseline detector, and engineer-reported decision confidence collected through the user-acceptance study described in Section VII.

It is also useful to distinguish between metrics that are directly measurable once the detection models are trained (mAP, precision, recall, inference time) and metrics that require a longer observation period in an operational deployment (reduction in mean time-to-repair, reduction in recurring root-cause incidents, and analytics-driven early detection of emerging line-level defect patterns). The former can be validated purely offline against the DeepPCB and FICS-PCB test splits described in Section V-C; the latter require the platform to be run over multiple production shifts so that the analytics and root-cause modules accumulate enough historical context to produce statistically meaningful trend and anomaly signals, which is why the evaluation protocol in this paper is presented as a design target and methodology rather than as a set of already-obtained empirical results.

X. DEPLOYMENT ARCHITECTURE

The platform is designed for containerized deployment using Docker, with separate containers for the React frontend, the Node.js/Express API gateway, the Python/FastAPI inference service, and the MongoDB database, orchestrated behind a reverse proxy. This separation allows the GPU-bound inference service to be scaled independently of the lightweight web tier, and allows the inference container to be deployed either on-premises, close to the inspection line for low-latency feedback, or on a cloud platform such as AWS or Render for organizations without local GPU infrastructure. A message-queue-based extension (e.g., using a lightweight broker between the API gateway and inference service) is identified as a natural enhancement to support multiple simultaneous inspection stations without blocking on synchronous inference calls, particularly as the number of connected cameras or upload stations grows.

XI. ETHICAL AND SOCIETAL CONSIDERATIONS

Automated inspection systems that also track operator- and shift-level statistics raise the possibility of misuse for punitive performance monitoring rather than process improvement. The design intentionally separates production-quality analytics (line, shift, and machine-level trends) from individually identifiable operator performance data, and restricts access to the latter to the Admin role, with anonymization applied before any data is used for model retraining, as noted in Section VIII. More broadly, by making AI decisions explainable through Grad-CAM visualizations and natural-language justifications, the system is intended to support rather than replace the judgment of quality-control engineers, keeping a human reviewer in the loop for any repair or rework decision rather than fully automating corrective action.

XII. DISCUSSION AND LIMITATIONS

The proposed architecture is deliberately scoped to software-level inspection and analytics; it does not include control of robotic pick-and-place machines, conveyor systems, or programmable logic controllers, and does not assume live synchronization with industrial camera rigs, all of which are identified as future work. The accuracy of the root-cause prediction and predictive-analytics modules is inherently dependent on the availability of sufficiently large and well-labeled historical production logs; in the absence of such data, these modules would need to operate in a heuristic or rule-based mode until sufficient training

data is accumulated through the continuous-learning feedback loop. Similarly, while Grad-CAM provides a widely accepted visual account of model attention, it approximates rather than guarantees a causal explanation of the network's decision, and the accompanying natural-language justification is generated from template reasoning over detected features rather than a fully independent causal model.

A. Choice of Detector Backbone

YOLOv11 was selected over alternative architectures such as Faster R-CNN, SSD, and EfficientDet primarily on the basis of inference speed relative to accuracy, which is a critical constraint for an inspection system intended to operate at production-line throughput rather than in an offline batch setting. Two-stage detectors such as Faster R-CNN typically achieve marginally higher localization accuracy on small, densely packed objects, which is relevant to fine-pitch PCB components, but at a substantially higher per-image inference cost that would limit the system's ability to keep pace with high-volume assembly lines. Single-stage detectors in the YOLO family have also demonstrated strong benchmark results specifically on DeepPCB and FICS-PCB in the studies reviewed in Section II, which further supports their use as the detection backbone for this platform, subject to confirmation through the empirical benchmarking proposed in Section XI.

B. Sample Explanation and Repair Templates

Table 8 illustrates representative explanation and repair-recommendation templates instantiated by the platform for two common defect classes. Each template is populated at run time with the specific component identity, location, and confidence score produced by the detection stage.

Table 8: Representative Explanation and Repair Templates

Defect Class	Explanation Template	Repair Template
Missing Component	Expected feature pattern absent at footprint; neighboring pads show no attachment.	Inspect feeder/nozzle; place component; re-run AOI.
Solder Bridge	Continuous solder region detected spanning two or more adjacent pads.	Rework joint with solder wick; verify stencil aperture; re-inspect.

These templates are intentionally simple and rule-driven in the current design so that every generated explanation remains traceable to an explicit, auditable rule; a natural extension, noted in Section XI, is to train a constrained language-generation model over a larger corpus of engineer-authored explanations to produce more varied and context-sensitive text while preserving traceability to the underlying detection and root-cause outputs.

XIII. FUTURE WORK

Several directions are identified for extending this work beyond the scope defined in Section III:

- **Empirical benchmarking.** Complete training and evaluation of the YOLOv11 component and defect detectors on DeepPCB, FICS-PCB, and the supplementary custom dataset, reporting the metrics defined in Table 7 against the baselines summarized in Table 1.
- **Root-cause validation.** Collection of real production-log data (feeder events, reflow-oven logs, stencil-cleaning schedules) to train and validate the XGBoost root-cause model against ground-truth maintenance records, rather than the illustrative feature schema described in Section IV.
- **Edge deployment.** Investigation of quantized or pruned detector variants (following the direction of edge-oriented detectors such as PCBDet [5]) to allow inference directly on inspection-line hardware without a persistent cloud connection.
- **Multi-modal inspection.** Extension of the defect-detection pipeline to incorporate additional imaging modalities, such as X-ray inspection for solder-joint voids or thermal imaging for early detection of overheating components, fused with the existing optical pipeline.
- **Federated continuous learning.** Enabling multiple manufacturing sites to contribute to the continuous-learning feedback loop described in Section IV-A without centrally sharing raw board images, using federated-learning techniques to preserve design confidentiality across sites.
- **Operator studies.** Conducting the user-acceptance evaluation outlined in Section VII with quality-control engineers to quantitatively measure the effect of explainability and repair recommendations on inspection time, rework rate, and decision confidence.
- **Physical integration.** Extending the platform toward live conveyor and pick-and-place integration, which is explicitly out of scope for the current software-level architecture but represents a natural progression toward closed-loop manufacturing control.

XIV. CONCLUSION

This paper presented SmartPCB-XAI, an integrated architecture for PCB inspection that extends beyond conventional component and defect detection to include explainable AI, root-cause prediction, automated repair recommendation, Digital Twin visualization, and predictive manufacturing analytics within a single Industry 4.0-oriented platform. Rather than proposing a new detection algorithm, the contribution of this work is the systematic design of an end-to-end pipeline that connects detection outputs to interpretable, actionable, and production-aware outcomes, addressing a gap identified across a broad review of recent PCB inspection literature summarized in Table 1 and Table 6. The architecture, module decomposition, dataset methodology, evaluation protocol, and deployment design presented in Sections IV through X provide a concrete and reproducible foundation for implementing and empirically validating an explainable, production-aware PCB inspection system, with the specific extensions outlined in Section XI defining a clear path from the current design to a fully validated, field-deployable platform.

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